

★ WEEK 2 · PROBLEM SET

Finance Applications: Three Hands-On Problems

Dollar Bars · CUSUM Events · Survivorship Bias

Each problem: Statement → Principles & Lab Guide → Hand Calculation → Claude Code Prompt

Small numbers you can compute with a pencil — then a metaprompt that lets Claude Code scale the same idea to real market data. Hand first, machine second.

Every Problem Follows the Same Four Steps

Work top to bottom — do not skip the hand calculation, it is where the understanding lives

1 · Statement

the question

A concrete finance scenario with a tiny dataset you can see at a glance.

one slide

2 · Principles

the tools

The formulas involved and a short lab guide: what to compute, in what order.

one slide

3 · Hand Calc

pencil first

Work every number by hand in a table.
Small n , clean arithmetic, no calculator fear.

1–2 slides

4 · Claude Code

scale it up

A metaprompt: paste it into Claude Code and it builds the full lab on real data.

one slide

Hand first, machine second — you check the machine, never the other way around.

Where the Data Comes From

Public sources first — the prompts fetch free data; CSV files are offline fallbacks only

Problem	Primary source (public · no signup)	Provided fallback file
P1 · Dollar bars	yfinance — real daily Close + Volume	fml_w2_prices_offline.csv
P2 · CUSUM events	yfinance — same series, log returns	fml_w2_prices_offline.csv
P3 · Survivorship	Simulated 500-fund universe (in-script, fixed seed)	fml_w2_p3_universe_sim.csv
Hand calculations	Printed on the slides (10 days · 8 returns · 6 funds)	fml_w2_p1_10days.csv · fml_w2_p3_funds.csv

Course rule: if data is free online (yfinance, FRED, OpenML), the prompt fetches it — files are for offline classrooms.

P1

PROBLEM 1 · DOLLAR BARS

Slice Ten Days by Value, Not by the Clock

Build dollar bars by hand and watch busy days earn more rows

Ten Days, Two Busy Spells, One Threshold

Earnings hit on days 3–4 and day 8. Slice these ten days into dollar bars with $\theta = \$30\text{M}$

Day	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Dollar volume (\$M)	8	6	20	22	5	4	6	25	5	6

The rule

- Walk left to right, adding up dollar volume.
- Close a bar the moment the running sum reaches \$30M, then reset to 0.
- A leftover sum at D10 stays an open stub.

Your deliverables

- The cumulative-sum table and every bar boundary.
- Bar count and calendar days per bar.
- One sentence: where did the market 'earn' extra rows — and why?

The Tools: Dollar Volume and a Threshold

One multiplication and one running sum — that is the whole machine

$$DV_t = P_t \cdot V_t \quad (\text{dollar volume of day } t)$$

$$\text{close bar } k \iff \sum_{t \in \text{bar } k} DV_t \geq \theta, \quad \theta = \$30\text{M (given)}$$

Reading the rule

- θ is the bar 'size' in traded value — the only knob.
- Busy days fill the bucket fast $\rightarrow$ more bars.

Lab guide — three steps

- Step 1: cumulative sum of DV, left to right.
- Step 2: mark each crossing of $\theta = 30$; reset.
- Step 3: count days per bar.

The Running Sum, Day by Day

Three crossings of $\theta = \$30\text{M}$ $\rightarrow$ three complete bars, one open stub

Day	DV (\$M)	Running sum	Action
D1 / D2 / D3	8 / 6 / 20	$8 \rightarrow 14 \rightarrow 34$	$34 \geq 30 \rightarrow$ close BAR 1 (D1–D3), reset
D4 / D5 / D6	22 / 5 / 4	$22 \rightarrow 27 \rightarrow 31$	$31 \geq 30 \rightarrow$ close BAR 2 (D4–D6), reset
D7 / D8	6 / 25	$6 \rightarrow 31$	$31 \geq 30 \rightarrow$ close BAR 3 (D7–D8), reset
D9 / D10	5 / 6	$5 \rightarrow 11$	stays open — no bar

10 time bars $\longrightarrow$ 3 dollar bars + 1 open stub

Busy Days Earn More Rows

Ten identical time bars become three activity-weighted dollar bars

3 + stub

Dollar bars from ten days ($\theta = \$30\text{M}$)

3 · 3 · 2

Calendar days inside bars 1, 2, 3

D8 = 25

One busy day nearly fills a whole bar alone

×2

Bar 3 forms twice as fast as bars 1–2 — information arrived faster

Read it with Week 2 language

- The clock gave every day one row; value gave rows where trading happened.
- Bar 3 is 'short' because day 8 was loud.

Why ML cares

- Each dollar bar carries a similar amount of information.
- That is one step closer to the IID rows models assume.

Scale It: From Ten Days to a Full Year

Paste the metaprompt — Claude Code builds real dollar bars and scores them

[INPUT] Prompt · Claude Code

```
Build a single Colab-ready Python script that:  
1) downloads 1 year of daily AAPL prices and volume  
   with yfinance (auto_adjust=True) and computes the  
   dollar-volume column  $DV = Close * Volume$ ;  
   (offline classroom: load fml_w2_prices_offline.csv)  
2) builds dollar bars with threshold = total DV / 50  
   using a running-sum loop, storing each bar's start  
   and end date and its log return;  
3) also computes plain daily (time-bar) log returns;  
4) prints a table: bar type | n bars | std | excess  
   kurtosis, and a histogram of days-per-dollar-bar;  
5) one-line verdict: which return series looks closer  
   to a bell curve, and why.
```

[OUTPUT] What to expect

```
~50 dollar bars; days per  
bar varies (2 - 15+)  
kurtosis: dollar-bar returns  
clearly lower than daily  
verdict: dollar bars closer  
to the bell curve
```

```
Same story as your 10-day  
table - just 25x more days.
```

P2

PROBLEM 2 · CUSUM EVENTS

Run the Smoke Alarm by Hand

Eight returns, one threshold — find every event the model would study

Eight Days Through a Symmetric CUSUM

Threshold $h = 3\%$. Which days ring the alarm — and in which direction?

Day	1	2	3	4	5	6	7	8
Log return r (%)	+1.0	+2.5	-1.0	+2.0	-0.5	-2.0	-2.5	+1.0

The rule (from the lecture)

- Track S^+ (up-drift) and S^- (down-drift) at the same time.
- Event when $S^+ > 3$ or $S^- < -3$; reset that side to 0.
- S^+ never goes below 0; S^- never above 0.

Your deliverables

- The full S^+ / S^- walk, day by day.
- Every event: day and direction.
- One sentence: what would change with $h = 2\%$?

The Tools: Two Running Sums and a Bell

The same two lines as the lecture — now you are the computer

$$S_t^+ = \max(0, S_{t-1}^+ + r_t) \quad S_t^- = \min(0, S_{t-1}^- + r_t)$$

$$\text{event} \iff S_t^+ > h \text{ or } S_t^- < -h \quad (\text{then reset that side to } 0)$$

Why two sums

- S^+ hears up-drift only; S^- hears down-drift only.
- A crash and a melt-up both matter — the alarm is two-sided.

Lab guide — the walk

- One row per day: update S^+ , update S^- , check both bells.
- After an event, only the triggered side resets.

The Full Walk: S^+ and S^- Side by Side

Two events in eight days — one up, one down

Day	r (%)	$S^+ = \max(0, S^+ + r)$	$S^- = \min(0, S^- + r)$	Alarm?
1	+1.0	1.0	0	—
2	+2.5	3.5 → EVENT, reset 0	0	EVENT ▲ up
3	−1.0	0	−1.0	—
4	+2.0	2.0	0	—
5	−0.5	1.5	−0.5	—
6	−2.0	0	−2.5	—
7	−2.5	0	−5.0 → EVENT, reset 0	EVENT ▼ down
8	+1.0	1.0	0	—

Check day 4: $S^- = \min(0, -1.0 + 2.0) = 0$ — the down-sum can only recover to zero.

Two Events in Eight Days

The alarm slept through six days of chatter and rang twice on real drift

$h = 3\% \Rightarrow 2 \text{ events in 8 days}$ (day 2 up-move, day 7 down-move)

What fired and why

- Day 2: two up-days in a row accumulated +3.5% — genuine up-drift.
- Day 7: three down-days accumulated -5.0% — a real slide, not one bad print.
- Single-day noise (day 3, day 5) never reached the bell.

Sensitivity: $h = 2\%$

- Day 2 still fires ($3.5 > 2$) — earlier, at the same spot.
- Day 6 now fires too: $S^- = -2.5 < -2$.
- Smaller $h \rightarrow$ more, smaller events — the Lab 2 trade-off you measured.

Scale It: From Eight Days to a Full Year

Paste the metaprompt — Claude Code sweeps the threshold and maps every event

[INPUT] Prompt · Claude Code

```
Build a single Colab-ready Python script that:  
1) downloads 1 year of daily AAPL prices with yfinance  
   and computes log returns;  
   (offline classroom: load fml_w2_prices_offline.csv)  
2) implements the symmetric CUSUM from class:  
    $S_{\text{pos}} = \max(0, S_{\text{pos}} + r)$ ,  $S_{\text{neg}} = \min(0, S_{\text{neg}} + r)$ ,  
   event when  $S_{\text{pos}} > h$  or  $S_{\text{neg}} < -h$ , reset that side;  
3) sweeps  $h = 0.5x$  to  $3x$  the return std (step  $0.5x$ ) and  
   plots event count vs  $h$ ;  
4) for  $h = 2x$  std, plots the price series with up-events  
   and down-events marked in different colors;  
5) two-line verdict: how the count decays with  $h$ , and  
   whether events cluster around news-heavy periods.
```

[OUTPUT] What to expect

```
event count falls roughly  
geometrically with  $h$   
 $h = 2 \text{ sigma}$ : a few dozen  
events in a year  
markers cluster in volatile  
stretches (earnings, macro)
```

```
Your 8-day walk, repeated  
250 times by the machine.
```

P3

PROBLEM 3 · SURVIVORSHIP

Resurrect the Missing Losers

Six funds, two deaths — measure how much the erased losers flatter the average

A Fund Database with Two Ghosts

Funds C and E died at the end of Year 1. Today's database shows only A, B, D, F

Fund	Year 1 return	Year 2 return	Status today
A	+10%	+6%	alive
B	+8%	+7%	alive
C	-30%	—	died end of Y1
D	+12%	+5%	alive
E	-25%	—	died end of Y1
F	+9%	+8%	alive

Your deliverables

- Year-1 average using only today's survivors.
- Year-1 average using ALL six funds (point-in-time).
- The survivorship bias in percentage points.
- One sentence: what would a naive backtest have concluded?

This is Week 2's Trap 1 with real arithmetic — the millionaires' survey, in miniature.

The Tool: One Subtraction

Bias = what the survivors say minus what actually happened

$$\text{Bias} = \bar{R}_{\text{survivors}} - \bar{R}_{\text{all funds}}$$

Point-in-time discipline

- To measure Year 1 honestly, use the funds that EXISTED in Year 1 — all six.
- Today's database silently deletes C and E.
- 'Who is missing?' is the first question for any dataset.

Lab guide — three steps

- Step 1: average Year-1 returns of A, B, D, F only.
- Step 2: average Year-1 returns of all six funds.
- Step 3: subtract — that gap is the bias.

Two Averages, One Ugly Truth

The survivors say +9.75%; the full universe says −2.67%

Universe	Funds included	Year-1 arithmetic	Average
Survivors only (today's DB)	A, B, D, F	$(10 + 8 + 12 + 9) / 4 = 39 / 4$	+9.75%
All funds (point-in-time)	A, B, C, D, E, F	$(10 + 8 - 30 + 12 - 25 + 9) / 6 = -16 / 6$	−2.67%

$$\bar{R}_{\text{surv}} = \frac{10+8+12+9}{4} = 9.75\% \quad \bar{R}_{\text{all}} = \frac{10+8-30+12-25+9}{6} = -2.67\%$$

$$\text{Bias} = 9.75\% - (-2.67\%) \approx 12.4 \text{ pp}$$

A 12-Point Mirage

Erasing two losers turned a losing year into a 'good' one

+9.75%

Year-1 average the
survivor-only
database reports

−2.67%

Year-1 average that
investors actually
experienced

12.4 pp

Survivorship bias —
the size of the mirage

2 / 6

Funds whose
disappearance
created it

The naive conclusion

- 'Funds beat cash by ~10% — buy any fund!'
- Real Year-1 investors, on average, lost money.

The Week 2 lesson

- Point-in-time data resurrects the losers.
- Backtests on survivor databases flatter every strategy.

Scale It: From Six Funds to Five Hundred

Paste the metaprompt — Claude Code simulates a fund universe and measures the mirage

[INPUT] Prompt · Claude Code

```
Build a single Colab-ready Python script that:  
1) simulates 500 funds x 24 monthly returns (Student-t  
   noise around fund-specific means, fixed seed);  
2) kills the 150 funds with the worst first-year  
   cumulative return at staggered dates in months  
   10 - 20 (their later returns become missing);  
   (offline classroom: load fml_w2_p3_universe_sim.csv)  
3) computes the first-year average return two ways:  
   (a) funds still alive at month 24 only,  
   (b) all funds that existed, point-in-time;  
4) prints both averages, the bias in pp, and plots the  
   two cumulative-average curves on one chart;  
5) two-line verdict: why the survivor curve is a lie,  
   and which real databases share this flaw.
```

[OUTPUT] What to expect

```
survivor-only 1y mean:  
  ~ +11%  
point-in-time 1y mean:  
  ~ +5%  
bias: ~ 6 - 7 pp  
  
Your 6-fund table, at  
database scale - the mirage  
does not average away.
```

Problem Set — Three Things Your Hands Now Know

You computed every number yourself; the code only repeats what you already understand

1

Dollar bars: busy days earn more rows

Three crossings of θ turned ten clock-days into three activity-weighted bars.

2

CUSUM: drift rings the bell, chatter does not

Two events in eight days — day 2's up-drift and day 7's slide; h sets the sensitivity.

3

Survivorship: erased losers flatter the average

+9.75% vs -2.67% — a 12.4 pp mirage from deleting just two dead funds.

Hand calculation first — then let Claude Code scale the same idea to real data.